



Predicting Academic Risk in College Students Using Enrollment & Performance Data

25FL-CS703-152W - APPLIED DATA
SCIENCE PROJECT

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PRESENTER

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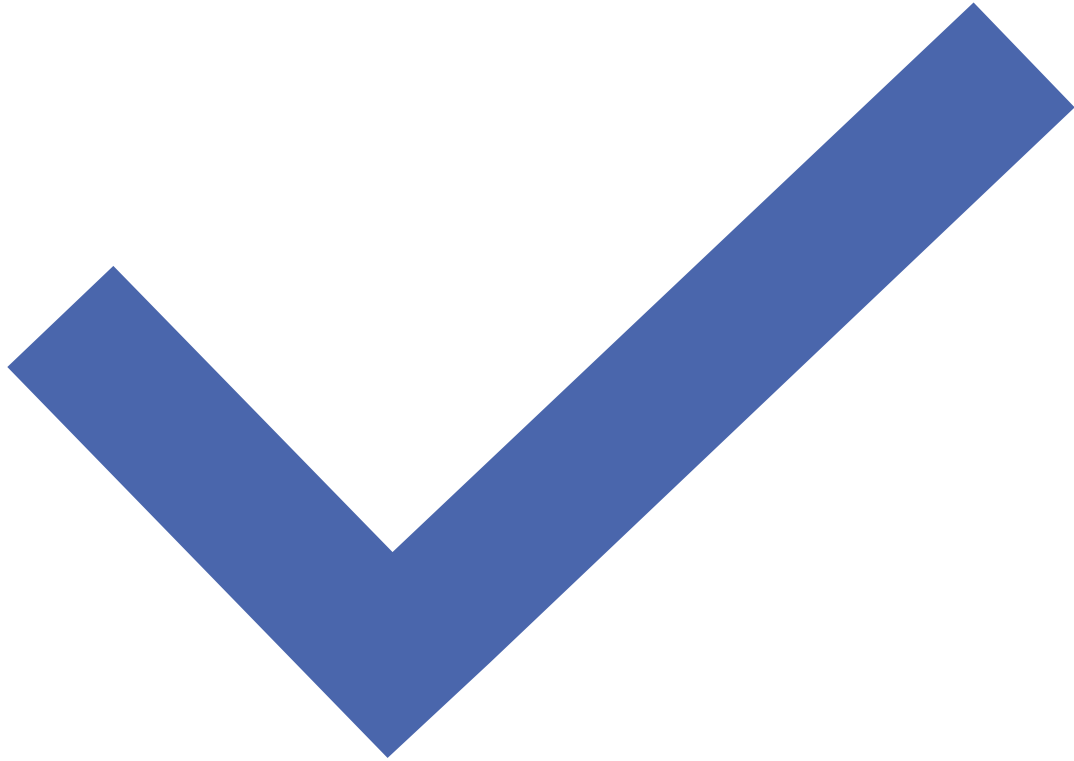


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Project Overview

- Business Issue:
 - The project addresses the challenge of identifying college students who may be at risk of academic failure. Institutions often struggle to detect early warning signs using large, dispersed datasets, leading to late or missed interventions.
- Business Goals:
 - Provide an early-alert approach for recognizing students who may require academic support.
 - Enable advisors and decision-makers to intervene earlier and allocate resources more effectively.



Project Overview Cont'd

- Data Mining Goals:
 - Develop a predictive model capable of estimating the likelihood that a student will be academically at risk.
 - Evaluate multiple modeling techniques to determine which provides the most accurate, reliable, and interpretable predictions.
- Data Mining Objectives:
 - Integrate institutional datasets (IPEDS + College Scorecard) into a single cleaned and prepared dataset.
 - Analyze data patterns, relationships, and risk indicators.
 - Build, validate, and evaluate predictive models using structured CRISP-DM methodology.
 - Select the strongest model and prepare it for deployment.



Summary of Each Project Phase

Phase 1.0 – Business Understanding

Purpose: Define the business problem and establish project expectations.

Key Components Completed:

- Identified the core business issue: predicting students who may be academically at risk.
- Defined business goals: support early intervention and improve academic outcomes.
- Defined data mining goals: build a predictive model to classify student risk.
- Established success criteria: accuracy, interpretability, and actionable outputs.
- Created the Project Plan outlining tasks, timeline, and required resources.



Phase 2.0 – Data Understanding

Purpose: Gather, review, and understand all project data.

Key Components Completed:

- Collected two datasets: IPEDS and College Scorecard.
- Described all key variables, structures, and data types.
- Explored the data using summary statistics and distribution checks.
- Identified initial patterns (e.g., variation in SAT scores, faculty salary, loan rates).
- Assessed data quality and documented issues such as missing values and duplicates.



Phase 3.0 – Data Preparation

Purpose: Prepare a clean, integrated dataset for modeling.

Key Components Completed:

- Selected relevant variables based on relevance to academic risk.
- Cleaned and transformed data (removed nulls, filtered NY institutions).
- Performed feature selection and ensured consistent formatting.
- Integrated datasets using UNITID to create one merged dataset.
- Finalized the modeling dataset used in Phase 4.0.



Phase 4.0 – Data Modeling

Purpose: Build and evaluate predictive models.

Key Components Completed:

- Selected two classification models: Logistic Regression and Decision Tree.
- Defined modeling assumptions, baseline parameters, and test design.
- Split the data into training/testing sets and applied 5-fold cross-validation.
- Trained both models and generated evaluation metrics.
- Compared performance using accuracy, precision, recall, F1-score, and confusion matrices.



Phase 5.0 – Modeling Evaluation

Purpose: Determine how well the models meet business and data mining goals.

Key Components Completed:

- Reviewed model results and confirmed they aligned with business expectations.
- Assessed model strengths, limitations, and interpretability.
- Determined that Logistic Regression best satisfied project needs.
- Verified modeling steps for correctness and consistency.
- Established next-step recommendations for deployment.



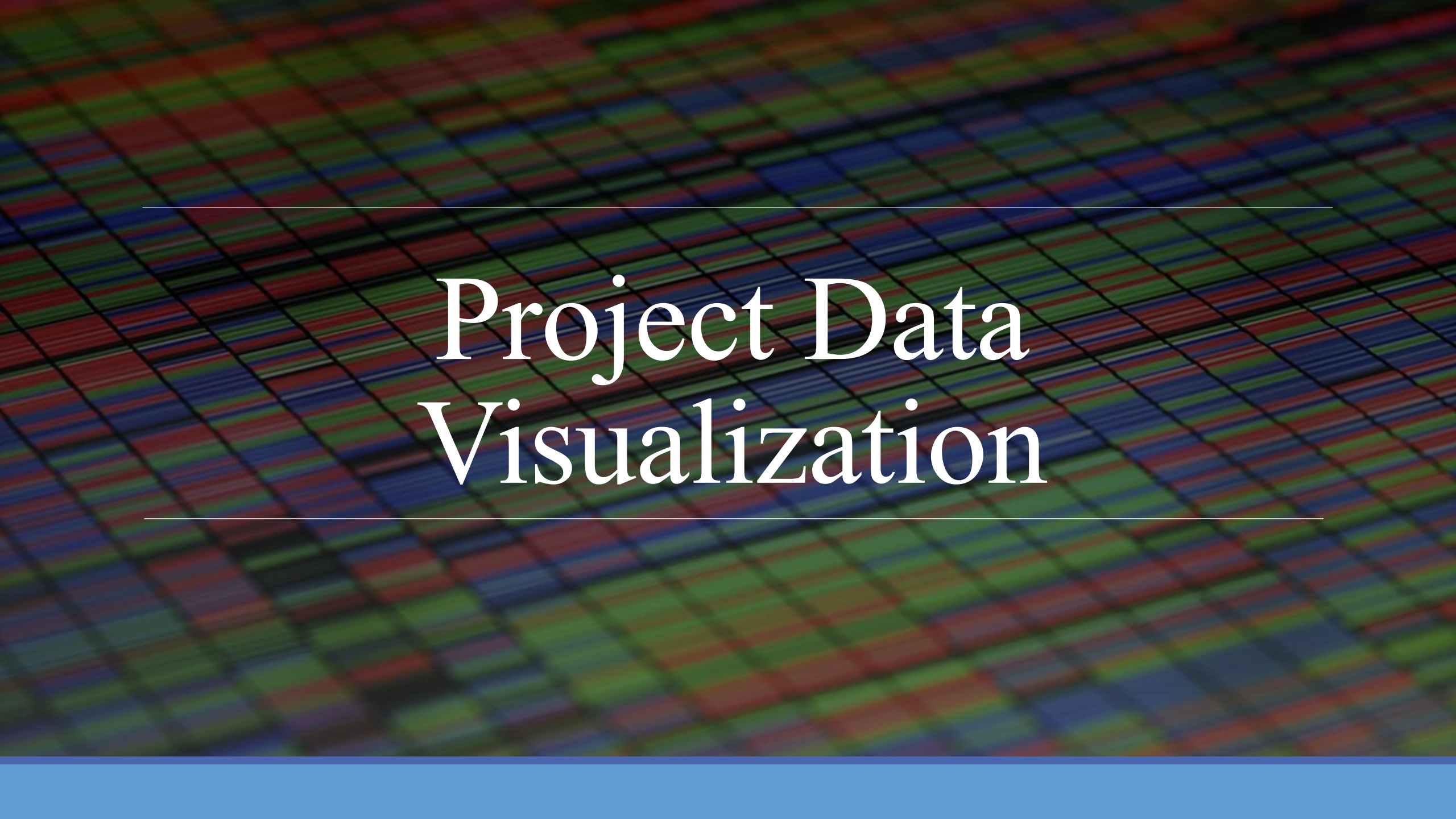
Phase 6.0 – Data Deployment

Purpose: Plan how the final model will be used in real operational settings.

Key Components Completed:

- Developed a Deployment Plan outlining how predictions will be generated each cycle.
- Developed a Monitoring & Maintenance Plan to track model performance over time.
- Completed the Team Experience Assessment reflecting on project workflow.





Project Data Visualization

Merged Dataset Info

Merged dataset (New York only) shape: (19193, 15)

	UNITID	INSTNM	STABBR	SAT_AVG	COSTT4_A	TUITIONFEE_IN	AVGFACSA	PCTFLOAN	STUFACR	CIPCODE	MAJORNUM	AWLEVEL	CTOTALT	CTOTALM	CTOTALW
0	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	38.0207	1	3	5	5	0
1	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	38.0207	1	5	0	0	0
2	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	99.0000	1	3	5	5	0
3	128586	Bais Binyomin Academy	NY	NaN	30640.0	9450.0	5250.0	0.0000	10.0	99.0000	1	5	0	0	0
4	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	11.0701	1	5	0	0	0
5	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	11.0803	1	3	0	0	0
6	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	14.0901	1	5	0	0	0
7	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	14.1099	1	5	5	3	2
8	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	14.1901	1	5	29	23	6
9	188340	Vaughn College of Aeronautics and Technology	NY	NaN	47174.0	28850.0	9242.0	0.5072	14.0	15.0201	1	3	0	0	0

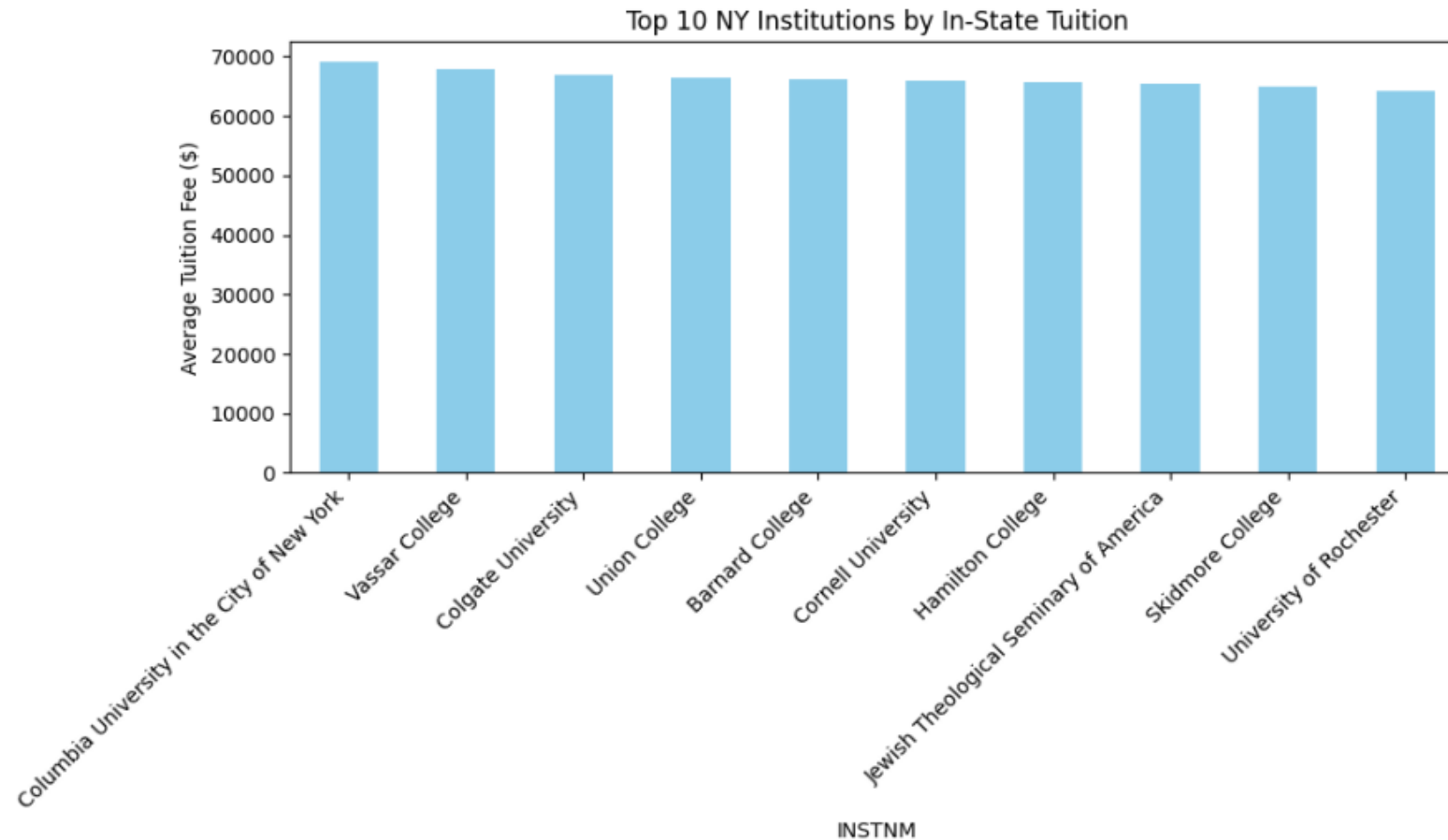
Statistical Summary

=== Scorecard Summary Statistics ===

	UNITID	ADM_RATE	SAT_AVG	UGDS	COSTT4_A	TUITIONFEE_IN	AVGFACSAL	RET_FT4	PCTFLOAN	STUFACR
count	6.429000e+03	1946.000000	1076.000000	5656.000000	3247.000000	3729.000000	3856.000000	2069.000000	5612.000000	5641.000000
mean	2.355496e+06	0.727594	1177.200743	2488.466054	30270.544195	17237.652722	7761.926608	0.727763	0.409069	15.285765
std	8.043861e+06	0.229527	144.813377	6157.338709	18375.779152	15644.112456	2853.086897	0.177499	0.274564	7.567888
min	1.006540e+05	0.000000	820.000000	0.000000	4274.000000	600.000000	488.000000	0.000000	0.000000	1.000000
25%	1.745700e+05	0.603975	1076.750000	116.000000	15765.000000	5688.000000	5946.750000	0.655000	0.152500	10.000000
50%	2.295400e+05	0.779150	1154.000000	494.000000	24517.000000	11790.000000	7409.000000	0.751400	0.442650	14.000000
75%	4.589550e+05	0.908175	1256.000000	2074.000000	40135.500000	23186.000000	9149.000000	0.840000	0.631300	19.000000
max	4.966450e+07	1.000000	1554.000000	156755.000000	87804.000000	69330.000000	25483.000000	1.000000	1.000000	191.000000

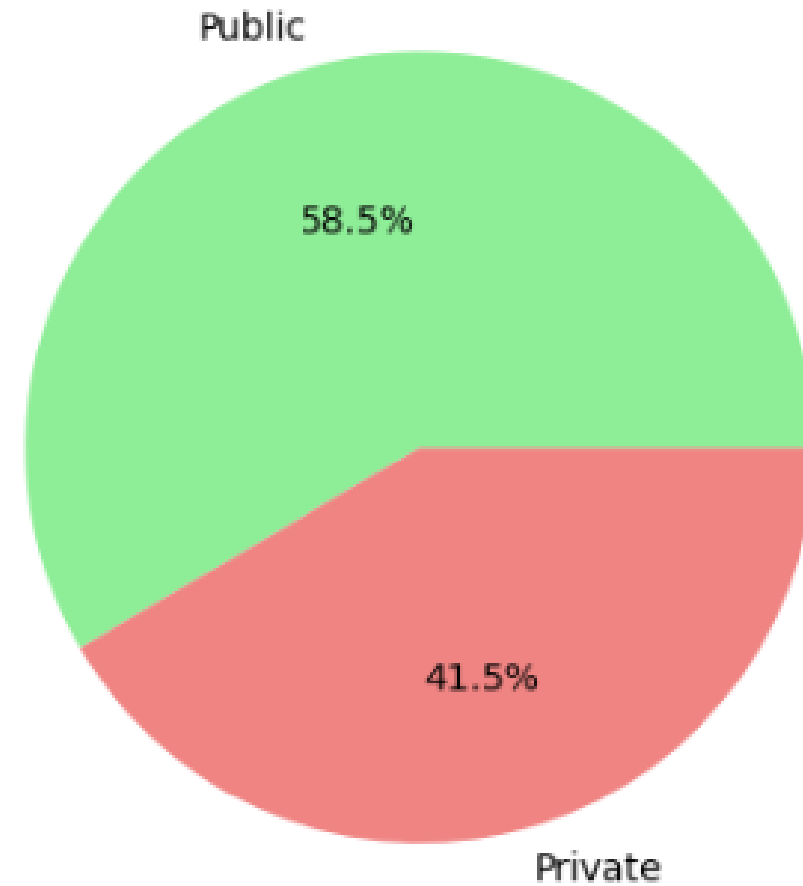
Top 10 Institutions by Tuition

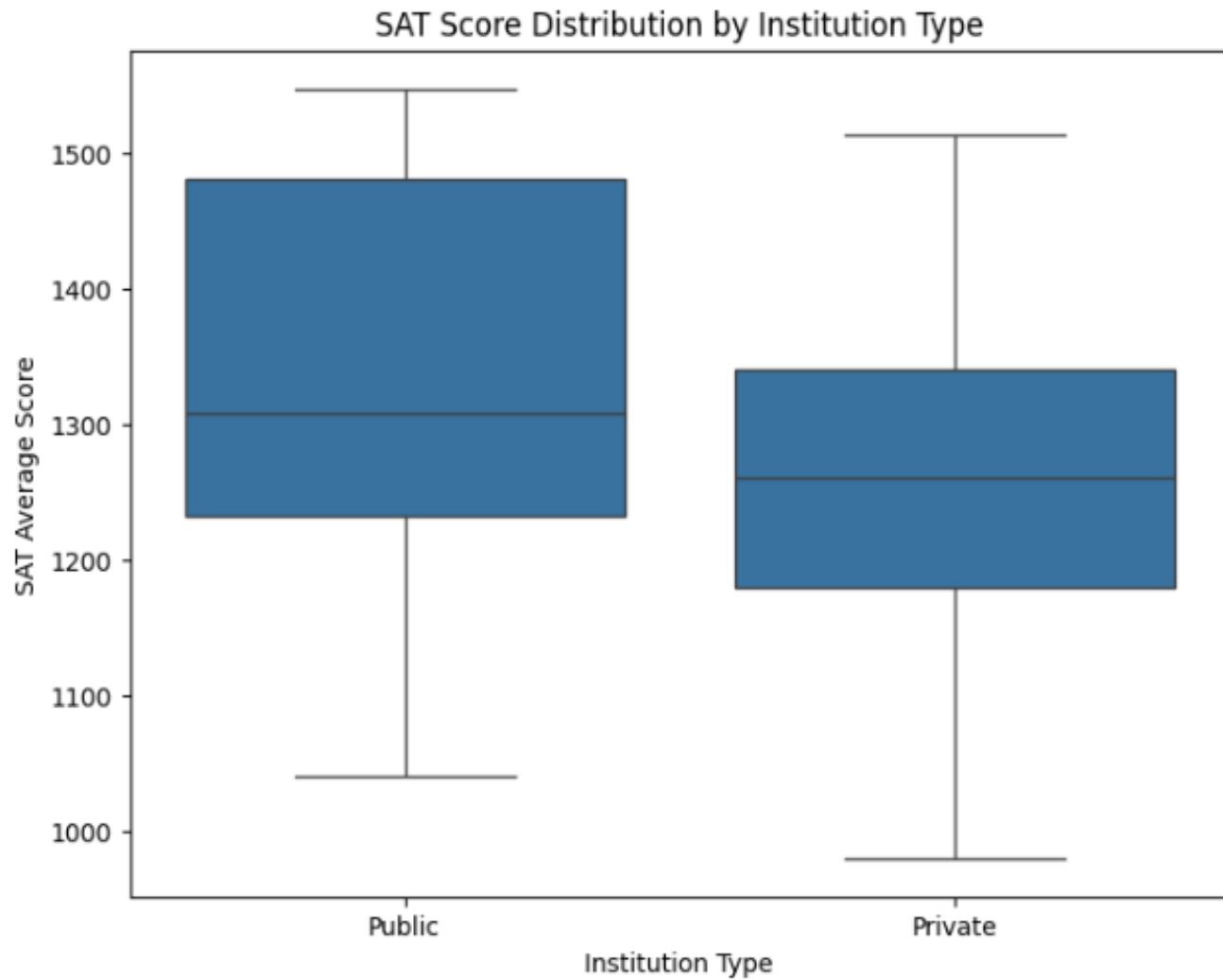
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Distribution of Public vs Private Colleges in NY

Distribution of Public vs Private Colleges in NY

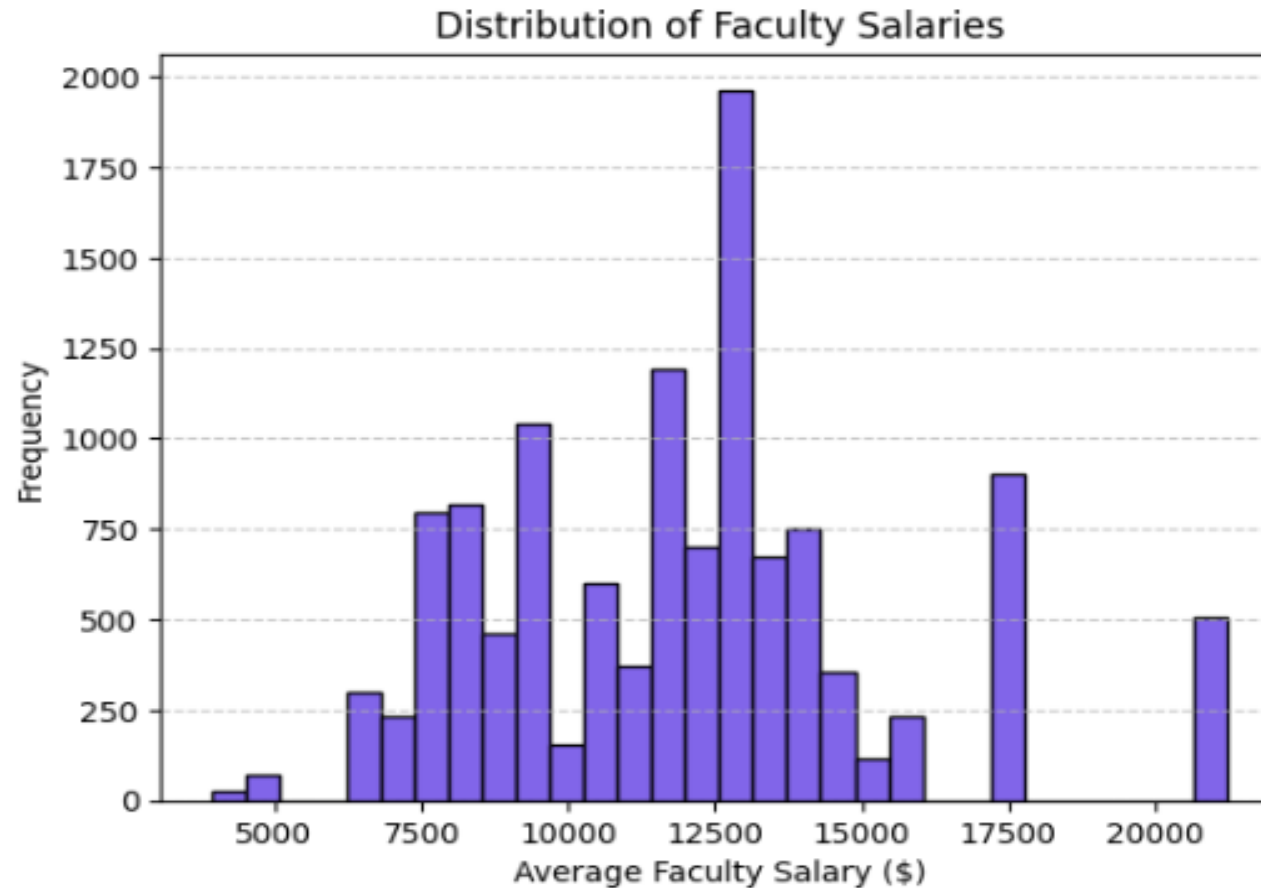




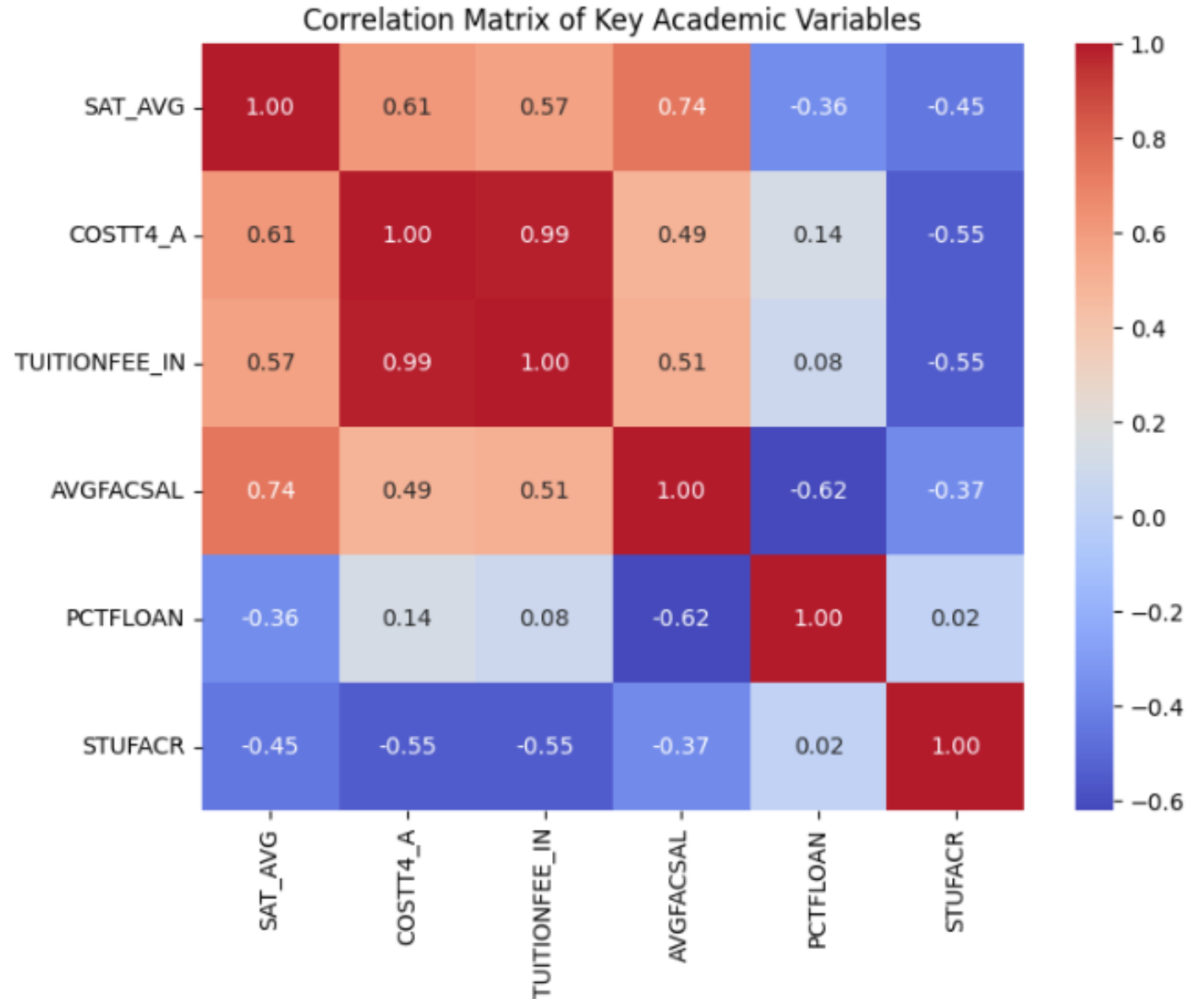
SAT Score
Distribution by
Institution Type

Distribution of Faculty Salaries

...



Correlation Between Variables



Models Used

Decision Tree

- Achieved 100% accuracy on test data.
- No misclassifications: perfect precision, recall, and F1-score for both classes.
- Confusion matrix showed 0 false positives / 0 false negatives.
- Simple yet highly interpretable model; performance suggests strong data separability.
- 5-fold Cross-Validation Accuracy: 1.000

Logistic Regression

- Also achieved 100% accuracy on test data.
- Classification report confirms perfect metrics across all performance indicators.
- Confusion matrix confirms ideal model behavior with no misclassifications.
- Offers a probabilistic perspective compared to the tree's rule-based splits.
- 5-fold Cross-Validation Accuracy: 1.000

```

=== Logistic Regression — Test Metrics ===
... Accuracy : 1.000
Precision: 1.000
Recall    : 1.000
F1-score  : 1.000

Classification report:
              precision    recall  f1-score   support

     0       1.00      1.00      1.00     3687
     1       1.00      1.00      1.00      158

 accuracy          1.00          1.00          1.00     3845
 macro avg         1.00          1.00          1.00     3845
weighted avg         1.00          1.00          1.00     3845

Confusion matrix:
[[3687  0]
 [  0 158]]

```

Logistic Regression — Confusion Matrix

Actual \ Predicted	0.00	1.00
0.00	3687	0
1.00	0	158

5-fold CV (Accuracy) — Decision Tree: 1.000 | Logistic Regression: 1.000

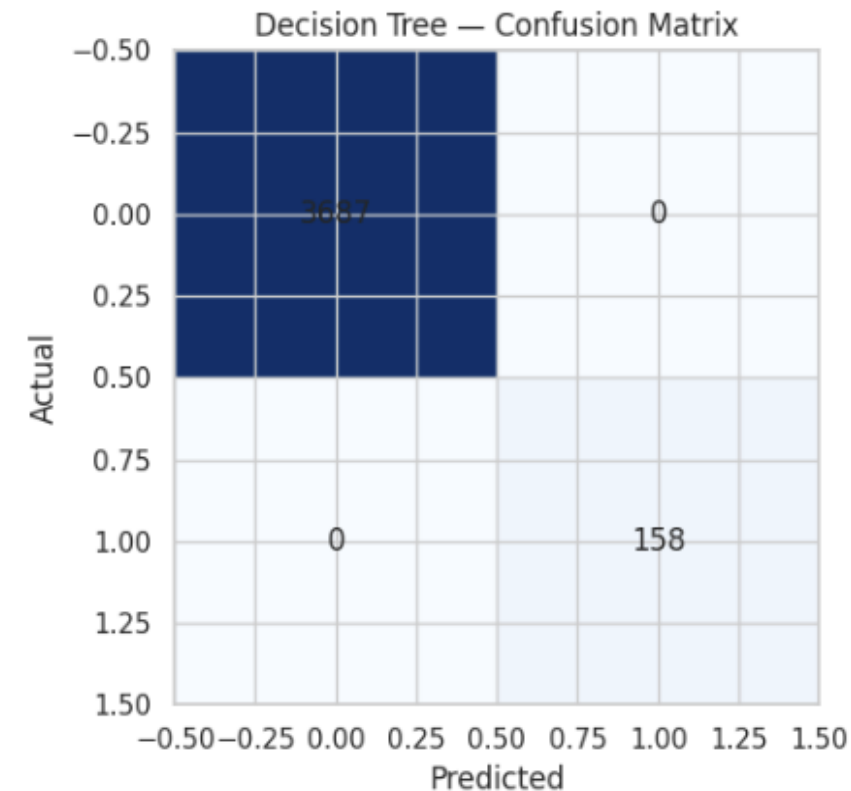
```
*** === Decision Tree - Test Metrics ===
Accuracy : 1.000
Precision: 1.000
Recall   : 1.000
F1-score : 1.000
```

```
Classification report:
              precision    recall  f1-score   support

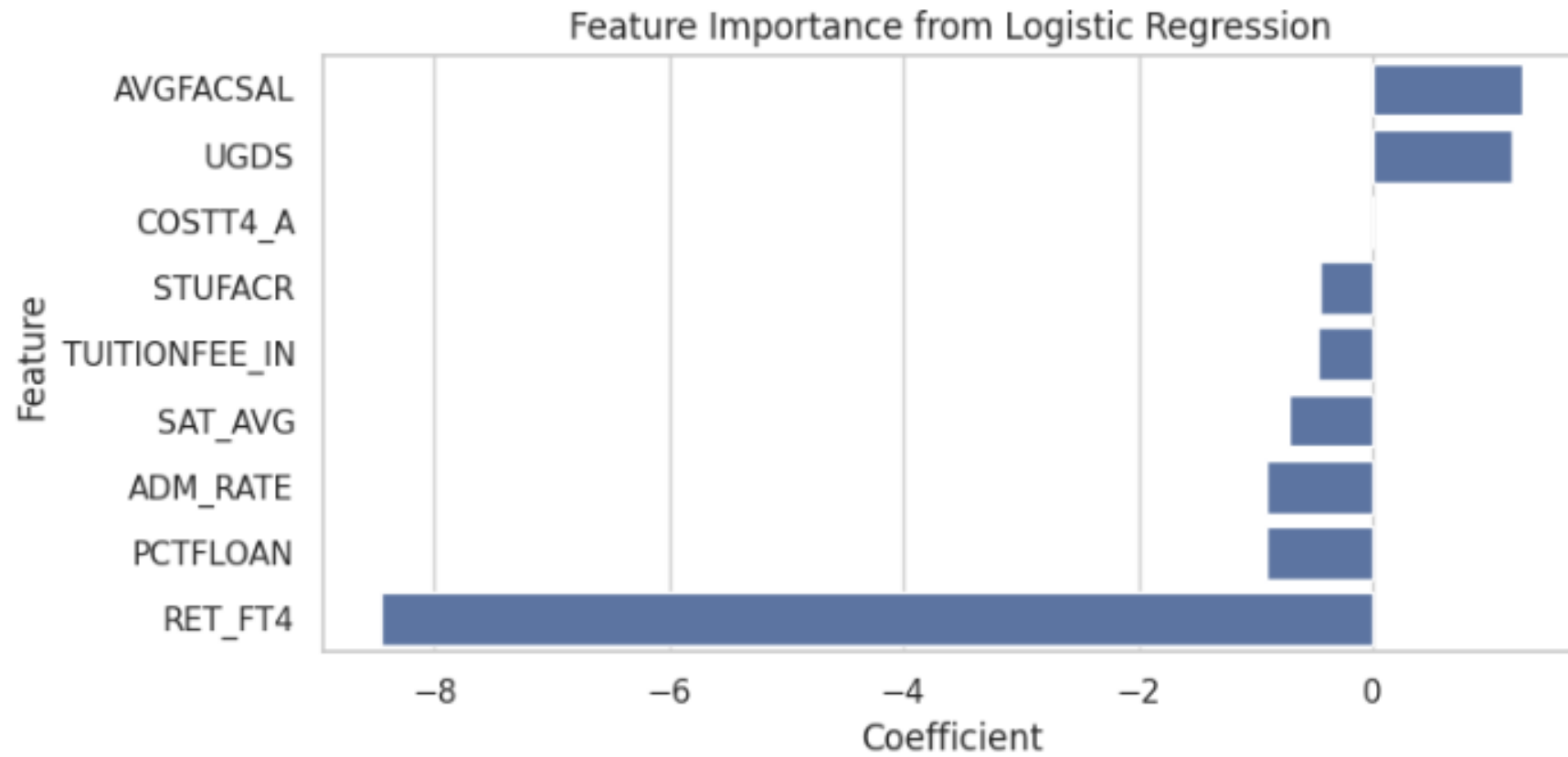
     0           1.00        1.00        1.00       3687
     1           1.00        1.00        1.00        158

 accuracy              1.00       3845
 macro avg           1.00        1.00        1.00       3845
 weighted avg        1.00        1.00        1.00       3845
```

```
Confusion matrix:
[[3687   0]
 [  0 158]]
```



Feature Importance



Model Eval & Business Impact

What the Model Shows:

- Model Type: Logistic Regression (binary classification)
- Target Variable: “At-Risk” status (1 = At Risk, 0 = Not At Risk)
- Predictors Included: Enrollment intensity, GPA, Pell Grant status, and tuition cost
- Key Output: Probability score between 0 and 1 for each student’s risk of dropping out
- Threshold Used: Students above 0.50 were flagged as “at-risk”

How It Relates to the Business Problem:

- The college needed a systematic way to predict which students were at risk of dropping out.
- This model uses existing data from enrollment and financial records to flag students early.
- It transforms raw data into actionable risk categories that academic advisors can use.

Model Eval & Business Impact Cont'd

Insights Gained:

- Students with part-time enrollment and lower GPAs were most likely to be flagged as “at-risk.”
- Receiving Pell Grants was a strong predictor, showing that financial aid recipients face greater retention challenges.
- Tuition levels influenced risk - students paying higher tuition without aid were less likely to be flagged.

Potential Business Impact:

- Allows early intervention planning (advising, tutoring, counseling) for at-risk students.
- Helps administrators allocate resources (like scholarships or mentorship) more effectively.
- Could improve overall retention rates and reduce loss of tuition revenue due to dropouts.

```
mirror_mod = modifier_ob.  
set mirror object to mirror.  
mirror_mod.mirror_object  
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False  
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True  
selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier_ob.  
mirror_ob.select = 0  
= bpy.context.selected_object  
data.objects[one.name].select  
print("please select exactly  
-- OPERATOR CLASSES --  
types.Operator):  
X mirror to the selected  
object.mirror_mirror_x"  
mirror X"  
context):  
context.active_object is not
```

LIVE Demonstration

Lessons Learned

Technology Lessons

- Learned how proper data cleaning and consistent preprocessing directly affect model accuracy.
- Gained experience comparing models using precision, recall, F1-score, confusion matrices, and cross-validation.
- Understood the need for monitoring input drift and prediction stability after deployment.

Management Lessons

- Saw how each CRISP-DM phase depends on the previous one and why strong early work prevents issues later.
- Learned the importance of aligning business goals with modeling decisions from the start.
- Recognized that clear documentation and version control are essential when producing multi-phase deliverables.

Personal Lessons

- Improved my ability to explain technical results in clear business terms.
- Developed stronger time-management skills working with weekly deadlines.
- Became more confident evaluating models objectively and selecting the most appropriate one.



Conclusion

- The project successfully followed all CRISP-DM phases and produced a fully evaluated, deployment-ready solution.
- The business and data mining goals were met: the model accurately identified at-risk students using integrated Scorecard + IPEDS data.
- Logistic Regression was selected after a detailed comparison because it performed strongly and remained interpretable, which is important for academic decision-making.
- Evaluation metrics and confusion matrix results confirmed the model's reliability, making it suitable for operational use.
- Deployment and monitoring plans were defined with clear tasks, timelines, performance checks, and retraining criteria.
- The project demonstrated how predictive analytics can support early intervention strategies and improve student success outcomes.



Thank You!
